

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
CO1 ASSESSMENT TOOL 2 – Agile Sprint Planning & User Story Workshop

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Code of Conduct

I I.Guru Hemanth Reddy (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Hemanth Reddy

Annexure A – Agile Sprint Planning & User Story Workshop

Smart College Bus Tracking System

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2 Executive Summary

The transition toward smart campus infrastructure has necessitated the development of intelligent transportation solutions. The Smart College Bus Tracking System aims to revolutionize the commuting experience for students, faculty, and administrators by providing real-time GPS tracking, accurate estimated times of arrival (ETAs), and a comprehensive fleet management dashboard. This document, titled “Annexure A – Agile Sprint Planning & User Story Workshop,” serves as the foundational Agile artifact for this project. It systematically breaks down the overarching project vision into manageable Epics and granular User Stories. By employing the Scrum framework, this document outlines the Product Backlog, establishes the priorities for Sprint 1, and details the acceptance criteria using the Behavior-Driven Development (BDD) Given-When-Then format. The estimations utilize the Fibonacci sequence to accurately reflect the complexity and effort required for each feature, ensuring a highly organized and scalable software development lifecycle.

3 Introduction

In modern educational institutions, managing a fleet of college buses poses significant logistical challenges. Students often experience prolonged wait times due to unpredictable traffic, while administrators struggle to monitor fuel efficiency, driver behavior, and route optimization. The Smart College Bus Tracking System is proposed as a cloud-based web and mobile application designed to alleviate these pain points.

This document outlines the rigorous Agile Sprint Planning and User Story Workshop conducted for the system. It adheres strictly to industry-standard software engineering practices, ensuring that all stakeholder requirements are accurately captured, estimated, and scheduled for development. By mapping out Epics, User Stories, and Acceptance Criteria, the engineering team is provided with a clear, unambiguous blueprint for development.

4 Agile Methodology Overview

Agile software development is an iterative and incremental approach that emphasizes flexibility, continuous improvement, and rapid delivery of functional software. Unlike traditional Waterfall methodologies, which rely on rigid sequential phases, Agile embraces changing requirements even late in the development cycle. In the context of the Smart College Bus Tracking System, Agile allows the development team to release core functionalities (such as basic live tracking) to the students early, while continuously refining complex administrative features (such as automated route optimization) in subsequent iterations based on real-world feedback.

5 Scrum Framework Explanation

Scrum is the specific Agile framework chosen for this project. It is designed to foster collaboration and accountability through predefined roles, artifacts, and ceremonies. * **Roles:** The Product Owner (representing the university administration and students), the Scrum Master (facilitating the process), and the Development Team. * **Artifacts:** The Product Backlog (the master list of all desired features), the Sprint Backlog (the specific features selected for the current 2-week cycle), and the Increment (the usable software produced). * **Ceremonies:** Sprint Planning, Daily Stand-ups, Sprint Review, and Sprint Retrospective. This document acts as the primary output of the initial Sprint Planning ceremony, establishing the direction for the first iteration.

6 Product Vision Statement

For students, faculty, and transport administrators who rely on the college's transportation network, the Smart College Bus Tracking System is a mobile and web-based application that provides real-time location tracking, automated notifications, and fleet analytics. Unlike the current manual scheduling system, our product leverages live GPS data and cloud computing to eliminate wait times, enhance student safety, and drastically reduce operational inefficiencies.

7 1. Project Vision and Stakeholders

7.1 1.1 Project Vision

To create a seamless, transparent, and highly efficient campus transportation ecosystem where every student and staff member can commute with absolute certainty regarding bus schedules, routes, and arrival times, while empowering the administration with actionable fleet analytics.

7.2 1.2 Objectives

- **Real-Time Visibility:** Provide live GPS tracking of all active college buses with an accuracy of under 10 meters and a refresh rate of 5 seconds.
- **User Convenience:** Deliver accurate ETAs and proximity push notifications to prevent students from waiting at stops during adverse weather.
- **Operational Efficiency:** Provide the transport department with analytics on route congestion, fuel consumption, and driver adherence to schedules.
- **Safety and Security:** Implement SOS features and secure driver authentication protocols.

7.3 1.3 Scope

In-Scope: Development of a cross-platform mobile application (iOS and Android) for students/faculty, a driver application for transmitting GPS data, and a web-based administrative dashboard. Integration with third-party mapping APIs (e.g., Google Maps) for routing and traffic data. **Out-of-Scope:** Hardware installation (the system assumes drivers will use standard GPS-enabled smartphones/tablets). Payment processing for bus fees is not included in phase one.

7.4 1.4 Stakeholder Identification

Identifying the correct stakeholders is paramount for capturing accurate requirements. The primary stakeholders include: 1. **Students and Faculty:** The end-users who commute daily. 2. **Bus Drivers:** The operators responsible for driving and transmitting location data. 3. **Transport Administrators:** The staff managing the fleet, assigning routes, and monitoring operations. 4. **University Management:** The executive sponsors funding the project. 5. **IT Support Team:** The personnel responsible for maintaining the system post-deployment.

7.5 1.5 Stakeholder Responsibilities and Table

Stakeholder Group	Primary Role	Key Responsibilities
Students Faculty	/ End User	Provide feedback on app usability; utilize the app to track buses.
Bus Drivers	Data Provider	Ensure the driver app is active during trips; follow assigned routes.
Transport Admins	System Manager	Assign drivers to routes; monitor fleet health; respond to delays.
University Mgmt	Sponsor	Approve budget and resources; review overall ROI and adoption metrics.
IT Support Team	Maintainer	Monitor server uptime; handle software updates and bug fixes.

8 Stakeholder Communication Plan

To ensure project success, a robust communication plan is established: * **Daily:** The development team will hold 15-minute Daily Scrums. * **Bi-Weekly:** Sprint Reviews will be conducted with Transport Administrators to demonstrate new features. * **Monthly:** Executive summaries and budget reports will be sent to University Management. * **Post-Deployment:** A feedback portal within the app will allow Students and Faculty to report bugs or request features directly to the Product Owner.

9 2. Epics and User Stories

To structure the development process, the system's requirements have been categorized into 6 major Epics, encompassing 20 detailed User Stories.

9.1 Epic 1: User Authentication and Profile Management

- **US1.1:** As a student, I want to register using my university email ID, so that I can securely access the tracking system.
- **US1.2:** As a driver, I want to log in using my assigned employee credentials, so that the system can verify my identity before I start a trip.
- **US1.3:** As an administrator, I want to manage user roles and permissions, so that unauthorized personnel cannot access the fleet management dashboard.

9.2 Epic 2: Real-Time GPS Tracking

- **US2.1:** As a student, I want to view a live map showing the current location of my assigned bus, so that I know exactly where it is.
- **US2.2:** As a driver, I want the app to automatically transmit my GPS coordinates every 5 seconds, so that the students receive accurate tracking data.
- **US2.3:** As an administrator, I want to view all active buses simultaneously on a master map, so that I have a macro-level view of the entire fleet.

9.3 Epic 3: Route and Schedule Management

- **US3.1:** As a student, I want to view the list of designated stops for my route, so that I can board the bus at the correct location.
- **US3.2:** As a student, I want to see the estimated time of arrival (ETA) for my specific stop, so that I can plan my departure from my home/hostel.

- **US3.3:** As an administrator, I want to create, edit, or delete bus routes, so that I can optimize paths based on changing traffic patterns or student demographics.
- **US3.4:** As a driver, I want to see my assigned route for the day upon logging in, so that I know which path to follow.

9.4 Epic 4: Notifications and Alerts

- **US4.1:** As a student, I want to receive a push notification when the bus is 2 kilometers away from my stop, so that I can reach the stop on time without waiting.
- **US4.2:** As a student, I want to be notified if my bus breaks down or is severely delayed, so that I can make alternative travel arrangements.
- **US4.3:** As a driver, I want a panic/SOS button in the app, so that I can instantly alert the administration in case of an emergency or accident.

9.5 Epic 5: Administrative Dashboard and Analytics

- **US5.1:** As an administrator, I want to generate reports on daily bus delays, so that I can identify systematically problematic routes.
- **US5.2:** As an administrator, I want to monitor the average speed of each bus, so that I can ensure drivers are adhering to safety regulations.
- **US5.3:** As an administrator, I want to view the attendance/occupancy metrics (if integrated with ID scanners), so that I can allocate larger buses to crowded routes.

9.6 Epic 6: System Settings and Support

- **US6.1:** As a student, I want to change the app's language settings, so that I can use the app in my preferred regional language.
 - **US6.2:** As a user (student/driver), I want access to a Help & Support section, so that I can report technical glitches or view FAQs.
 - **US6.3:** As an administrator, I want to configure the global notification distance thresholds, so that alerts are sent at appropriate times based on city traffic conditions.
 - **US6.4:** As a student, I want to save my home stop as a "Favorite", so that the app opens directly to my relevant route information.
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10 3. Acceptance Criteria

Acceptance criteria define the exact conditions under which a User Story is considered "Done." They are written using the industry-standard Behavior-Driven Development (BDD) format: **Given [Context], When [Action], Then [Result]**.

10.0.1 US1.1: Student Registration

- **Given** a student is on the registration screen,
- **When** they enter a valid university email (e.g., @simats.edu) and a secure password,
- **Then** the system creates an account and sends a verification email.
- **Given** a student enters a non-university email (e.g., @gmail.com),
- **When** they click register,
- **Then** the system displays an error message stating "Only university emails are permitted."

10.0.2 US1.2: Driver Login

- **Given** a driver is on the login screen,
- **When** they enter correct employee credentials,
- **Then** they are redirected to the Driver Dashboard.

10.0.3 US1.3: Admin Role Management

- **Given** an admin is in the user management portal,
- **When** they select a user and change their role to ‘Driver’,
- **Then** the system updates the database and grants the user driver-app access.

10.0.4 US2.1: View Live Map

- **Given** a student has selected Route A,
- **When** they open the map view,
- **Then** a bus icon is displayed at the latest coordinates received from the driver’s device, updating every 5 seconds.

10.0.5 US2.2: GPS Transmission

- **Given** the driver app is in an active trip state,
- **When** 5 seconds elapse,
- **Then** the app sends a POST request with the current lat/long to the server backend.

10.0.6 US2.3: Admin Master Map

- **Given** an admin is logged into the web dashboard,
- **When** they navigate to the Fleet View,
- **Then** they see multiple bus icons representing all active trips simultaneously.

10.0.7 US3.1: View Route Stops

- **Given** a student is viewing Route A,
- **When** they click “View Stops”,
- **Then** a chronological list of all designated stops is displayed.

10.0.8 US3.2: View ETA

- **Given** a student has selected their specific stop on the map,
- **When** the system calculates the distance from the bus to the stop,
- **Then** an ETA is displayed based on current traffic conditions via the Maps API.

10.0.9 US3.3: Edit Routes

- **Given** an admin is in the Route Configuration menu,
- **When** they drag and drop a waypoint to a new street,
- **Then** the system saves the new route polyline and updates all assigned drivers.

10.0.10 US3.4: Driver Route View

- **Given** a driver has just logged in,
- **When** they view their daily schedule,
- **Then** the app displays their assigned Route ID and a start button.

10.0.11 US4.1: Proximity Notification

- **Given** a student has subscribed to notifications for Stop X,
- **When** the bus enters a 2-kilometer radius of Stop X,
- **Then** a push notification is triggered to the student’s phone.

10.0.12 US4.2: Delay Alert

- **Given** a bus is immobilized for more than 15 minutes,
- **When** the admin flags the route as “Delayed”,
- **Then** all students assigned to that route receive an urgent alert.

10.0.13 US4.3: SOS Button

- **Given** a driver is in an active trip,
- **When** they hold the SOS button for 3 seconds,

- **Then** a high-priority alert with the exact coordinates is sent to the Admin Dashboard.
- 10.0.14 **US5.1: Delay Reports**
- **Given** an admin is on the Analytics page,
 - **When** they select “Generate Monthly Delay Report”,
 - **Then** the system exports a CSV file detailing all trips that arrived late by more than 10 minutes.
- 10.0.15 **US5.2: Speed Monitoring**
- **Given** a driver exceeds 60 km/h,
 - **When** the GPS data is processed,
 - **Then** a speeding violation is logged in the driver’s personnel file.
- 10.0.16 **US5.3: Occupancy Metrics**
- **Given** the bus is equipped with an RFID scanner,
 - **When** a student taps their ID,
 - **Then** the system increments the current occupancy count by 1.
- 10.0.17 **US6.1: Language Settings**
- **Given** a student is in the Settings menu,
 - **When** they select ‘Tamil’ from the language dropdown,
 - **Then** all UI elements immediately translate to Tamil.
- 10.0.18 **US6.2: Help and Support**
- **Given** a user clicks “Support”,
 - **When** they submit a ticket,
 - **Then** the ticket is routed to the IT Support Team queue.
- 10.0.19 **US6.3: Configure Thresholds**
- **Given** an admin is in Global Settings,
 - **When** they change the notification threshold from 2km to 3km,
 - **Then** all future proximity alerts are triggered at the 3km mark.
- 10.0.20 **US6.4: Save Favorite Stop**
- **Given** a student views a specific stop,
 - **When** they tap the ‘Star’ icon,
 - **Then** that stop is pinned to their home screen for quick access.

11 4. Product Backlog

The Product Backlog is a prioritized master list of all features. Priority is determined by Business Value and dependencies.

PB ID	Epic	User Story	Priority	Business Value	Dependencies	Story Points
PB-01	Epic 1	US1.1: Student Registration	High	9/10	None	3
PB-02	Epic 1	US1.2: Driver Login	High	9/10	None	2
PB-03	Epic 1	US1.3: Admin Role Management	Medium	7/10	PB-01	3
PB-04	Epic 2	US2.2: GPS Transmission (Driver)	High	10/10	PB-02	8

PB ID	Epic	User Story	Priority	Business Value	Dependencies	Story Points
PB-05	Epic 2	US2.1: View Live Map (Student)	High	10/10	PB-04	13
PB-06	Epic 2	US2.3: Admin Master Map	High	8/10	PB-04	8
PB-07	Epic 3	US3.3: Edit Routes (Admin)	High	8/10	None	5
PB-08	Epic 3	US3.1: View Route Stops (Student)	Medium	7/10	PB-07	3
PB-09	Epic 3	US3.4: Driver Route View	Medium	7/10	PB-07	3
PB-10	Epic 3	US3.2: View ETA	High	9/10	PB-05, PB-08	8
PB-11	Epic 4	US4.1: Proximity Notification	Medium	8/10	PB-10	5
PB-12	Epic 4	US4.3: SOS Button (Driver)	High	10/10	PB-02	3
PB-13	Epic 4	US4.2: Delay Alert	Medium	6/10	None	3
PB-14	Epic 5	US5.1: Delay Reports	Low	5/10	PB-10	5
PB-15	Epic 5	US5.2: Speed Monitoring	Low	6/10	PB-04	5
PB-16	Epic 5	US5.3: Occupancy Metrics	Low	4/10	Hardware	8
PB-17	Epic 6	US6.1: Language Settings	Low	3/10	None	2
PB-18	Epic 6	US6.4: Save Favorite Stop	Low	4/10	PB-08	2
PB-19	Epic 6	US6.2: Help and Support	Low	3/10	None	2
PB-20	Epic 6	US6.3: Configure Thresholds	Low	4/10	PB-11	3

12 5. Sprint Backlog

Sprint 1 Plan * Sprint Goal: Establish the foundational architecture, user authentication, and the core ability for a driver app to transmit GPS data to the cloud. *** Sprint Duration:** 2 Weeks

Task ID	Selected User Story	Development Tasks	Task Owner	Est. Hours	Status
SP1-01	US1.1: Student Reg.	Set up Firebase Auth; Design UI; API Integration.	Frontend Dev 1	12	In Progress

Task ID	Selected Story	User	Development Tasks	Task Owner	Est. Hours	Status
SP1-02	US1.2: Login	Driver	Create secure login portal; Connect to employee DB.	Backend Dev	8	Not Started
SP1-03	US2.2: Transmit	GPS	Implement background location services in mobile app.	Mobile Dev	16	Not Started
SP1-04	US2.2: Transmit	GPS	Create WebSocket server to receive live coordinates.	Backend Dev	20	Not Started
SP1-05	US1.3: Roles	Admin	Create basic admin web portal UI and role toggles.	Frontend Dev 2	10	Not Started

13 6. Story Point Estimation

The project utilizes the **Fibonacci Sequence (1, 2, 3, 5, 8, 13, 21)** for Agile estimation. This non-linear scale prevents false precision and accurately reflects the inherent uncertainty in complex software tasks.

13.0.1 Estimation Justifications

Story Point	Justification and Example
1 Point	Trivial tasks with zero complexity. E.g., changing a text label or updating a color scheme.
2 Points	Simple tasks with known implementation steps. E.g., US1.2 (Driver Login) . Standard API call, no complex logic.
3 Points	Moderate tasks requiring some UI/UX and backend connection. E.g., US1.1 (Student Registration) . Requires validation and email verification.
5 Points	Complex tasks requiring logic or minor third-party integrations. E.g., US4.1 (Proximity Notification) . Requires calculating radial distances periodically.
8 Points	Highly complex tasks involving core infrastructure or significant unknowns. E.g., US2.2 (GPS Transmission) . Requires handling mobile background states, battery optimization, and WebSockets.
13 Points	Massive tasks encompassing UI, complex backend, and heavy API usage. E.g., US2.1 (View Live Map) . Requires rendering custom markers, smoothing GPS jitter, and integrating Google Maps SDK.

14 7. Sprint Goal and Deliverables

14.1 Sprint Goal

The objective of Sprint 1 is to deliver a functional Minimum Viable Product (MVP) core where users can register, log in, and the driver application can successfully ping its location to the cloud database in real-time.

14.2 Features Delivered

By the end of Sprint 1, the following features will be operational: 1. Complete Authentication system for Students and Drivers. 2. Backend server capable of establishing WebSocket connections. 3. Driver mobile application capable of capturing and transmitting GPS data in the background.

14.3 Expected Outcomes

The primary outcome is validating the technical feasibility of real-time GPS tracking using the chosen tech stack. Successfully completing this sprint proves the architecture can handle continuous data streams.

14.4 Sprint Review

A formal meeting will be held at the end of the sprint where the development team will demonstrate the Driver App transmitting coordinates to a raw database log in front of the Transport Administrators.

14.5 Sprint Retrospective

Following the review, the Scrum team will discuss: * *What went well*: E.g., Firebase auth integration was seamless. * *What went wrong*: E.g., iOS background location permissions caused delays. * *Actionable improvements*: Allocate more time for OS-specific permission handling in future sprints.

14.6 Definition of Done (DoD)

A User Story is only considered “Done” when: 1. The code is written, peer-reviewed, and merged into the main branch. 2. Unit tests are written and passing. 3. The feature meets all defined Acceptance Criteria. 4. The feature is deployed to the staging environment and verified by QA.

15 Risk Assessment Table

Risk	Probability	Impact	Mitigation Strategy
GPS Signal Loss in Tunnels	High	Medium	Implement data caching in the driver app to transmit data once signal is restored.
Battery Drain on Driver App	High	High	Optimize location sampling rates and require drivers to keep devices plugged into the bus charger.
Third-Party API Rate Limits	Medium	High	Implement local distance calculations (Haversine) to reduce calls to Google Maps API.
Low Student Adoption	Low	Medium	Conduct a campus-wide awareness campaign and offer initial incentives.

16 Assumptions and Constraints

16.1 Assumptions

- All college buses are equipped with a functioning 12V charging port for driver devices.
- Drivers possess a basic understanding of operating a smartphone application.
- The university campus and surrounding transit routes have adequate 4G/5G cellular coverage.

16.2 Constraints

- **Time:** The initial MVP must be deployed before the commencement of the new academic semester (12 weeks).
- **Budget:** Premium third-party routing APIs (like Google Maps Enterprise) must be used judiciously to avoid exceeding monthly cloud expenditure limits.
- **Technology:** The system must be compliant with local data privacy laws, ensuring student location data is not permanently stored or shared.

17 Sprint Timeline (Week-wise)

- **Week 1 (Sprint 1):** Backend Architecture setup, Database Schema creation, API Gateway initialization.

- **Week 2 (Sprint 1):** Authentication modules complete, Driver GPS transmission module built and tested.
- **Week 3 (Sprint 2):** Student UI map integration, WebSocket connections established for live receiving.
- **Week 4 (Sprint 2):** Admin dashboard map view, basic route creation tools.
- **Week 5 (Sprint 3):** ETA calculation engine, proximity notification logic.
- **Week 6 (Sprint 3):** QA Testing, User Acceptance Testing (UAT) with a sample group of 5 buses.

18 Conclusion

The Agile Sprint Planning and User Story Workshop for the Smart College Bus Tracking System establishes a robust, highly structured foundation for development. By meticulously breaking down the project into 6 Epics and 20 User Stories, and rigorously defining acceptance criteria and story point estimations, the engineering team is equipped with a clear roadmap. The adoption of the Scrum framework ensures that the project remains flexible, transparent, and focused on delivering maximum business value to the university, ultimately resolving the long-standing challenges of campus transportation logistics.

19 References

[1] K. Schwaber and J. Sutherland, *The Scrum Guide*. Scrum.org, 2020. [2] R. S. Pressman, *Software Engineering: A Practitioner's Approach*, 8th ed. McGraw-Hill, 2014. [3] I. Sommerville, *Software Engineering*, 10th ed. Pearson, 2015. [4] M. Cohn, *User Stories Applied: For Agile Software Development*. Addison-Wesley, 2004. [5] K. Rubin, *Essential Scrum: A Practical Guide to the Most Popular Agile Process*. Addison-Wesley, 2012.